

Creating and Delivering Transferable Object-Based Light Content

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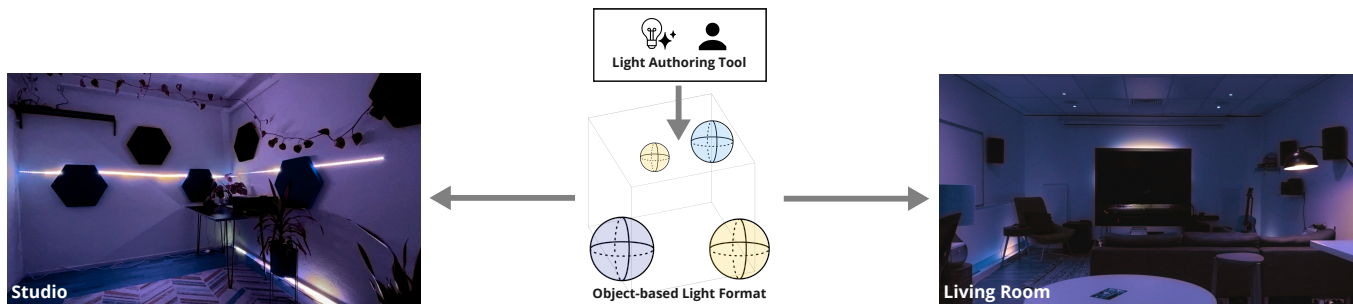


Figure 1: We enable the design of transferable light content by authoring with an abstract object-based format and then flexibly rendering across arbitrary playback environments. This figure depicts two endpoints: a studio and a living room, where the same object-based content is being rendered. In this example, the studio was used to author content that was later delivered to the living room endpoint without the author having any knowledge of the latter.

Abstract

Light cues and ambient lighting are a well-established tool for interaction designers with broad application across smart homes, human-robot interaction, entertainment, and others. However, light content is often designed for specific endpoints, i.e. fixture layouts, making the process of transferring designs across endpoints challenging. To address this, we explore how light content can be authored in an abstract object-based format that maintains design intent, whilst being transferable across endpoints. We present an authoring tool for object-based lighting, and a flexible lightscape renderer to reproduce this content on arbitrary light fixture combinations. In an application example, we create a light design commonly found in concert venues and demonstrate how it can be both authored and reproduced on consumer devices in a studio and a home environment. We highlight challenges and opportunities around transferability and discuss how transferable light content may support researchers and practitioners in creating more reproducible and longer-lived designs.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **HCI design and evaluation methods**; **Interaction paradigms**; **Interaction design process and methods**; **Systems and tools for interaction design**.

Keywords

Light Design, Transferability, Authoring Tools, Flexible Rendering

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1 Introduction

Light cues and ambient lighting, which we collectively refer to as *lightscares*, have long been used by human computer interface (HCI) researchers and professionals as communication tools across application contexts like internet-of-things, robotics, entertainment, smart home, and automotive. Lighting cues are used to convey basic information like a device's on/off state, to structure user interaction in the case of conversational agents listening to speech input, or to alert a user to relevant external events or objects in the automotive context, where cues often contain additional positional information. They are also used to convey more abstract notions such as a robot's attractiveness [20], being kept in the thoughts of a remote loved one

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[7], or as part of a multimodal immersive experiential environments [24].

A shared characteristic across these deployments is that cues are designed for specific light fixture positions or arrangements, meaning design work is strongly linked with, or even locked into, the hardware it was created for. This may well be appropriate when considering one-off deployments for applications involving simple lighting systems. However, as both designs and fixture distributions become more complex, two key limitations emerge. From a researcher's perspective, *reproducibility* becomes a challenge. If light design work cannot be transferred from one deployment to another, it instead needs to be re-created, which requires both time resources and a thoroughly documented reference to be available. This friction alone can make it wholly unfeasible. From a practitioner's perspective, the same constraints can lead to *short-lived designs* with a limited audience. One-off deployments then place a limit on the time and effort put into the design, leading to potentially less refined or polished results. It can also hinder the development of more general best practices or design systems that can be applied across design artifacts.

If we instead imagine lighting content in a more traditional media format that can be distributed to and then reproduced across various devices, such as audio or video, these limitations become less pronounced. A set of light cues designed for a social robot [5, 11, 21], for example, may be deployed to any other robot that has light fixtures. A set of internal or external light signals for an autonomous vehicle [10, 17, 23] may be deployed to different models of the same manufacturer, or to any other vehicles.

In light of these considerations, we set out to explore what a system comprising both the authoring and rendering of transferable light content may look like. The remainder of this paper is structured as follows. Section 2 provides a high-level overview of the diverse ways HCI researchers deploy lighting. Section 3 introduces the notion of transferable object-based light content and maps out the benefits of and requirements for transferable lighting design. Section 4 details our development of an authoring tool and a flexible lightscape renderer (FLR) and is followed up by an application example in Section 5. Finally, we discuss challenges, opportunities and future work around transferable light content.

2 Related Work

To provide appropriate context for this research, we will provide a brief overview of two areas of particular relevance: the use of light across the HCI community, and how light content is created and deployed outside academia.

Lighting has been part of a HCI designer's tool set since the first light emitting diode (LED) on a circuit board. Even within such constraints, designers have explored ways to expressively convey information [9] and recent work by Kunchay and Abdullah [14] highlights that light as a communication modality, even in constrained use cases like smart speakers, there remains significant potential to improve the intuition of lighting cue designs. Other works consider the use of ambient lighting beyond a single LED. Hoggenmueller et al. [12] explored the use of low-resolution lighting displays as a versatile and aesthetic medium for ambient interfaces and proposed design recommendations based on a

deployment study. Hommes and Kipp [13] connected distributed smart lights (Philips Hue) across two homes to convey a sense of physical closeness between separated individuals. Gaver and Gaver [7] also explored mediating connectedness using connected but physically remote light bulbs.

While lighting in the automotive space has a long history, it has seen recent interest among researchers exploring how autonomous vehicles (AVs) communicate with humans. The absence of a driver creates a need for new ways to communicate with pedestrians. Al-Taie et al. [1], for example, used LED strips on the outside of an AV to negotiate space-sharing with cyclists [1]. Wilbrink et al. [23] explored the use of a light-based internal human-machine-interface to increase trust in a turning scenario with notable success [23]. Hensch et al. [10] deployed light-based communication externally to communicate with surrounding pedestrians, reporting how cues were perceived as partially trustworthy but generally unintuitive, and highlighting the need for further exploration of the modality [10].

In the area of human-robot interaction, light has been used as a mechanism to convey information and affect social perceptions of robots. Terada et al. [21], for example, used a small personal computer and micro-controllers to control LEDs to explore light and color cues for conveying emotion [21]. Albastaki et al. [2] used the Processing software environment [6] to create light cues for a virtual robot [2]. Hoggenmueller et al. [11] covered a robot with Raspberry Pi-controlled LED panels to create a mobile low-resolution display and subsequently deployed it in an urban environment to investigate interactions with bystanders [11]. Ginosar and Cauchard [8] propose an expressive light system using an LED strip embedded on a commercial quadcopter. A shared characteristic of all these works is the deployment of custom-made designs for a specific light fixture layout.

In industry, lighting design is present in a wide range of contexts, including concert venues, exhibitions, and retail. These can differ widely in complexity. Consider, for example, a retail location that may have one scene of static ambient lighting per store, while a theater play may involve different lighting for each of its 20 scenes. Concert lighting is notable in this context, as lighting can be tightly synced to musical events, leading to highly animated and complex designs. As a result, various lighting technologies have been developed for this use case. Light shows are designed by dedicated professionals who either work at a specific venue, or accompany an artist to realize their shows in each venue along a tour. Shows are often designed ahead of time through computer-aided design and pre-visualisation using software such as WYSIWYG [4], Vectorworks Vision [22] and Capture [3]. A designer's site-specific work then involves adapting the content to the available fixtures through programming with tools like GrandMA [18]. Each venue will feature custom fixture setups with custom coded content which are structured in a channel-based format where authoring is applied to individual fixtures or groups thereof. When cues need to be transferred from one venue to another, this requires either a perfectly matching fixture layout, or for the show to be re-programmed to fit a new setup [19].

3 Towards Transferable Object-based Light Content

The above examples of light content design are authored in a bespoke manner for each endpoint which limits the reproduction of the designs to endpoints which are identical to the system on which they were authored. However, if there were a way to separate authoring and reproduction through abstraction without compromising the creative intent of the designer, light content would be transferable to a variety of endpoints that need not be identical. This would facilitate, for example, concert lighting to be reproduced on smart lights in the home, or light cues emitted by coordinated drones to be reproduced on integrated lighting systems in a vehicle.

This concept of transferability is not trivial, and requires two key components as denoted in Figure 1. First, an abstract representation is required for intuitively encapsulating creative intent. Object-based light content is one such abstract representation, which will be described in the subsequent sections. Second, a renderer is required to interpret the abstract representation and based on the location and characteristics of the available light fixtures reproduce the lightscape experience at the endpoint.

Separating design work from endpoints shares commonalities with approaches in the audio space. Object-based audio is well-established in gaming, although rendering is mostly restricted to standardized endpoints like headphones or stereo speakers. A recent flexible approach to audio rendering is the Dolby Atmos Flex-Connect [16] system which enables flexible rendering of the Dolby Atmos [15] audio format to arbitrary layouts in the living room. However, the analogs between flexible audio and light rendering are severely limited. As a result, the technologies and systems required to implement and deliver transferable object-based light content must be developed.

4 Implementation

4.1 Authoring Object-Based Light Content

The authoring of object-based light content aims to capture author intent without space or light fixture dependencies. An overview of the tool we developed for this purpose is shown in Figure 2.

Light objects are represented by a spatial primitive in 3D space which is mapped to the playback environment. A discrete spatial object is defined as a shape which activates lights that fall within it and its feather volume. The FLR supports a range of parametric shapes such as ellipsoids, cuboids, bi-cones and cylinders. These are parameterized, not exhaustively, by their 3D position, attitude (orientation), size, and color which can be altered over time using keyframes and curves in this authoring tool. It is also possible to modulate parameters at render-time using external inputs to the renderer. Specifying the position and size within an abstract 3D space allows an author to generalize the position and extent of a sequence without prior knowledge of the final playback environment.

More abstract area-effect objects allow parametric functions, and data such as tensors, images or videos, to be sampled before being combined with an activation function, which may have discrete spatial object behavior, in order to be rendered. An example of this is the *slice* effect, which separates an object into active and

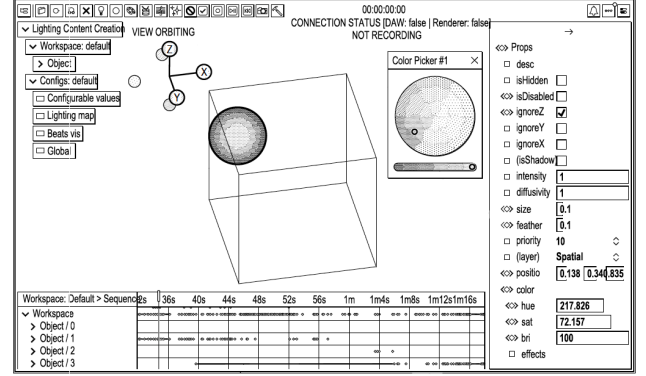


Figure 2: User interface of our authoring tool. Key components are a 3D visualization (center), parameter controls for each object (right), and a keyframe timeline (bottom). Whilst there are multiple objects visible in the timeline, we only show a single object in the 3D visualization for clarity.

inactive segments by propagating a parametric wavefront across the endpoint. It uses a discrete spatial activation function to determine which light fixtures are driven by the effect whilst the slice function itself determines their activation values and colors based on a 4D space-time function evaluated at the light fixture positions. The slice effect has additional parameters such as phase center, duty cycle, dithering, smoothing, temporal frequencies, and spatial frequencies which are on normalized scales across the 3D space they are created in.

4.2 Flexible Rendering of Light Content

Light content, comprised of objects and associated metadata, is combined with user preferences and endpoint configuration data, such as light fixture positions and capabilities, by the FLR to render a lightscape flexibly. A high-level diagram showing the signal flow of an FLR is shown in Figure 3. Some key components of the rendering process are described, not exhaustively for brevity, in this section.

4.2.1 Object Activations. The first stage of the lightscape rendering pipeline involves determining object-fixture activation values on every layer. Layers are described in more detail later and are used to segment objects. At this stage, the tensors $\mathbf{A} \in \mathbb{R}^{N_l \times N_o \times N_f}$ and $\mathbf{C} \in \mathbb{R}^{N_l \times N_o \times N_f \times 3}$ containing the activation values and RGB colors respectively are constructed. N_l is the number of layers, N_o is the number of objects, and N_f is the number of light fixtures.

Discrete spatial object activation functions test if a light fixture falls within its volume and sets the corresponding element of $\mathbf{A}$ to 1 or, if it falls within the feathering region of the object, computes an activation between 0 and 1. If there is no intersection of the object and fixture, the activation value is 0. For example, the spherical object activation function between the i^{th} object and j^{th} light fixture on the l^{th} layer can be written as

$$A_{l,i,j} = \begin{cases} 1 & \text{if } d_{i,j} \leq s_i \\ (s_i + f_i - d_{i,j}) / (f_i + \epsilon) & \text{else if } d_{i,j} \leq s_i + f_i \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

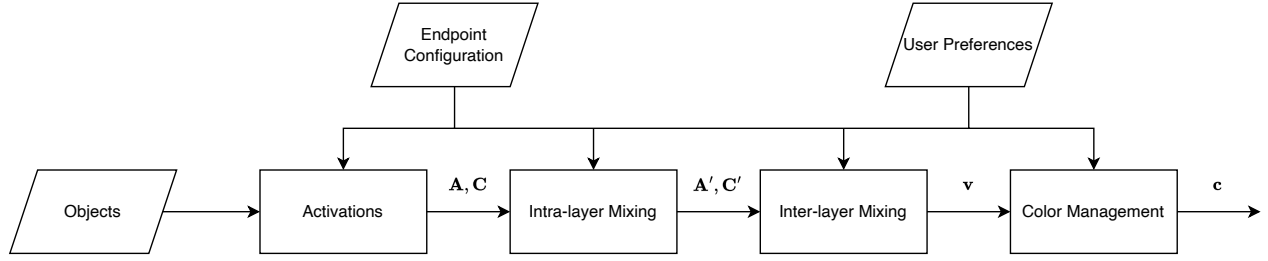


Figure 3: A high-level signal flow diagram of a flexible lightscape renderer.

where $d_{i,j}$ is the Euclidean distance between the i^{th} object and j^{th} light fixture, s_i is the radius of the object's sphere and f_i is the object's radial feather size. The FLR also supports the notion of extended lights where the object-fixture activation function, e.g. (1), considers the light fixture as an extended volume rather than a point.

4.2.2 Layers and Mixing. Layers provide a pseudo-priority scheme for the author as the inter-layer blending stage is dependent upon the order of the layers. Authors place higher priority objects on higher layers in a similar fashion to how images are composited from multiple layers. The current implementation of the renderer has five layers which are, in order from top to bottom, the overlay, spatial 2, spatial 1, ambient object and ambient bed layers. Any type of object may be played on any layer. The exception to this is ambient bed objects, that we do not detail in this paper, and the ambient bed layer, which are exclusive to one-another. The intra-layer mixing process produces the matrices $A' \in \mathbb{R}^{N_l \times N_f}$ and $C' \in \mathbb{R}^{N_l \times N_f \times 3}$ containing activation values and RGB colors respectively. This is a mixing of objects on a per-layer basis resulting in an activation vector and color vector for every layer. All layers, with the exception of the overlay and ambient object layers, utilize activation-weighted dodge mixing of colors. The overlay layer takes the activation and color corresponding to the maximum activation value across all object-fixture activations for each fixture in the intra-layer mixing process. This enables the creator to place high priority light objects on the overlay layer, which may have functional purposes, and not have their meaning altered due to the mixing process. The inter-layer mixing process is a-over-b compositing and requires an alpha parameter. This is computed during the intra-layer mixing process using the activation values and the intensity parameter of the object. The inter-layer mixing process produces the output color vector, $v \in \mathbb{R}^{N_f \times 3}$, containing a color setpoint for each light. The combined intra- and inter-layer mixing process is illustrated in Figure 4.

4.2.3 Configuration Generation. The input to the endpoint configuration generator contains the size and shape of the endpoint, the positions and characteristics of the light fixtures, and information about the location of the user(s). Light positions normalized by the room scale and shape are used to compute object-light activations, scale time-frequencies and spatial-frequencies of objects, and potentially scale the brightness of codewords sent to particular fixtures. The endpoint configuration also contains data to transform

the light-object positions into alternate domains whenever meta-data indicates the light-object is positioned relative to a certain landmark, such as a screen, within the endpoint. Moreover, the configuration also contains the information required by the color management module, depicted in Figure 3, which transforms the renderer output color vector, v , into a vector of codewords, c , to be sent to the light fixtures in the endpoint. This module is responsible for ensuring an arbitrary set of light fixtures will display consistent colors in addition to providing a global and local brightness control mechanism.

4.2.4 Personalization. The current implementation of the FLR allows the user to personalize the lightscape experience by way of some basic controls. The first of these is the control of the perceptual brightness levels output by the renderer which is achieved by scaling and/or clipping, depending on the configuration, brightness during the color management process. We also expose a control which fades in objects tagged as ambient-only whilst simultaneously fading out objects which are tagged as immersive-only. By default, objects are tagged as neither and this control does not affect objects with this default setting. However, if an author chooses to utilize this feature in the metadata, the user is then able to control the immersive-ness of the experience. During the configuration generation process, light fixtures are also assigned to zones. One use of this zone identification is the ability for these personalization controls to be applied on a zonal basis across the endpoint.

5 Application Example: Bringing Concert Lighting into the Home

To demonstrate the flexible lightscape concepts and technologies, we now apply them to a use case: designing a lightscape to accompany a live music performance which also transfers to light fixtures available in the home. While this use case is less functional than many targeted HCI applications, it is useful to evaluate our approach for both the creative palette it offers to designers, and how well designs transfer to other endpoints. Light content that accompanies music has a diverse range of design goals, ranging from accompanying specific musical accents with precisely synced cues, to more symbolic representations of mood or narrative content. An example of stage lighting fulfilling both of these functions can be found [here](#).

Using this example as a general reference, we created a short snippet of music and designed lighting to accompany it. We chose

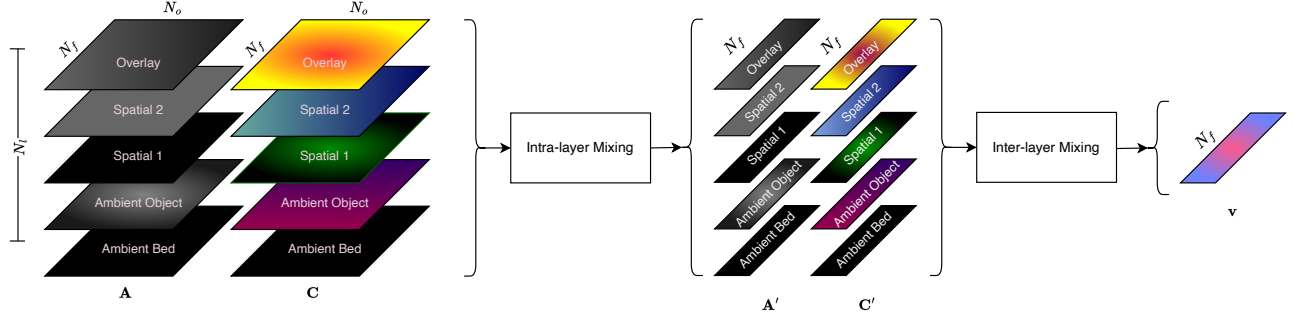


Figure 4: A visualization of the mixing process throughout the renderer. Note that the colors shown are for illustration and are not representative of the actual mixing processes.

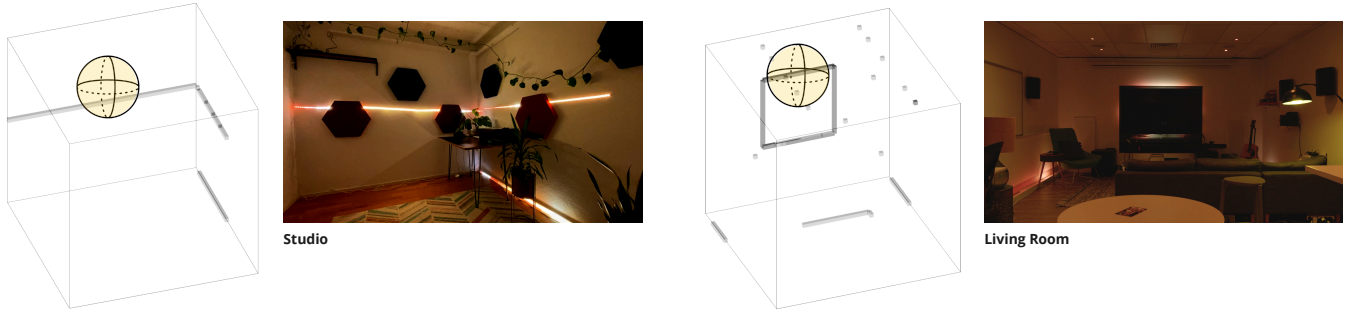


Figure 5: Rendering a lightscape to LifX LED smart lights in two endpoint environments: a studio (56 addressable LED strip segments) during authoring and a living room (88 addressable LED strip segments, 13 down lights, and one floor lamp) for later playback. The content was authored without any specific consideration being made for the living room. We only show one light object in the 3D visualization for clarity.

a warm color palette and designed cues for the core musical elements, including drums, cymbals, keys, and brass. These were then distributed spatially, either positioned at specific locations or across the entire room depending on their underlying musical characteristics. Authoring took place in a studio environment, shown in the left of Figure 5. The 3D layout shows the positions of three consumer LED strips, which could be used for monitoring to evaluate design choices during authoring. The right of Figure 5 shows the fixture distribution in a living room equipped with various smart lights. A video illustrating the abstract object representation and the resulting lightscape in both the authoring and the living room environments is provided in the supplementary materials of this paper. It should be highlighted that the two environments are on different continents, and that the third author (of this paper) who designed the lighting in the studio environment did not experience their design in the living room environment before the recording was made.

6 Discussion

The ability for light experiences to transfer to arbitrary endpoints is the main value proposition of the abstract object-based format and flexible rendering technology we propose. When considering the video accompanying Section 5, we can reasonably claim that the

design successfully transferred between the two locations without any manual intervention, despite the significant differences in the number of fixtures (56 to 102) and their positions shown in Figure 5. Elements like the front center snare, the kick drum covering the entire floor, and room-wide effects were all retained and color differences in the video are solely due to the recording equipment used. However, once the number of fixtures, their positions, and/or capabilities begin to differ more extensively, or an author uses predominantly small, localized objects as opposed to room-wide effects, challenges may emerge. In the given example, the central snare drum cannot be reproduced in rooms that have no fixtures at that location without distorting the position and/or shape of the object. The FLR can augment the position of an object if an object fails to activate a sufficient amount of light fixtures. The author opts in to this functionality by configuring the object metadata and potentially trades-off spatial fidelity to ensure an object is rendered in the endpoint regardless of the loss of fidelity.

Transferability is both complex and complicated as it is affected by (i) the format and rendering algorithms, (ii) the endpoint the content was authored in, (iii) the endpoint the experience is transferred to, and, crucially, (iv) the extent to which the author utilizes the features of the format and its metadata. As a result, we cannot, in general, expect experiences to have guaranteed transferability. One

way to address this is to provide authors with methods and tools that help them assess the transferability of their designs *during* authoring and thereby supporting them in utilizing the format to its fullest potential. To date, we have relied on subjective evaluation to iterate on the FLR algorithms, however, we see the utility of developing a quantitative transferability metric to (i) predict how transferable a particular experience is, and (ii) evaluate how successfully an experience has transferred to an endpoint. Measuring the transferability of an experience is useful for the development of more capable flexible rendering algorithms and future work will therefore focus on developing frameworks for estimating and measuring transferability. This will provide authors with a feedback mechanism they can use when constructing their metadata, and identifying endpoint configurations that are particularly well-suited for reproducing transferable light content.

The scalable nature of object-based light content offers several key advantages for both researchers and practitioners. By abstracting the lighting design away from specific layouts and hardware configurations, we enable greater reuse of designs across different endpoints. For researchers, this scalability facilitates more reproducible studies, in endpoints where this was previously not feasible, as lighting conditions can be automatically reproduced across diverse playback environment and light fixtures. It also allows for longitudinal studies where lighting designs can evolve over time without being constrained by initial hardware choices. For practitioners, scalable object-based content opens up new possibilities for creating lighting experiences that can adapt to a wide range of environments - from large-scale public installations to individual smart home setups. This adaptability not only extends the lifespan and reach of lighting designs but also encourages the development of more sophisticated and nuanced lighting experiences.

7 Conclusion

In this work, we explored the challenges and opportunities around light content that can be transferred beyond the fixture setup it was originally designed on. We introduced object-based light content, a flexible lightscape renderer and authoring tool that enables the design of lighting that can be transferred to arbitrary endpoints automatically. The flexibility of the format and the renderer is key to enabling this automatic transfer, and we presented an application example that illustrates how a design, authored in a studio, can be reproduced in a separate living room with a significantly different location and number of light fixtures. However, there is significant work remaining to estimate and optimize the transferability of a design, especially for situations where authoring and playback environments widely differ. Additionally, we are engaging with lighting design professionals to evaluate how well our system (i) enables them to realize their design goals and (ii) preserves the same when reproducing their designs in unknown environments.

References

- [1] Ammar Al-Taie, Graham Wilson, Euan Freeman, Frank Pollick, and Stephen Anthony Brewster. 2024. Light it Up: Evaluating Versatile Autonomous Vehicle-Cyclist External Human-Machine Interfaces. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (New York, NY, USA, 2024) (CHI '24). Association for Computing Machinery. doi:10.1145/3613904.3642019 event-place: Honolulu, HI, USA.
- [2] Almohannad Albastaki, Marius Hoggemueller, Frederic Anthony Robinson, and Luke Hespanhol. 2020. Augmenting Remote Interviews through Virtual Experience Prototypes. In *32nd Australian Conference on Human-Computer Interaction (OzCHI '20)*, Sydney, NSW, Australia. ACM, New York, NY, USA, 9. doi:10.1145/3441000.3441057
- [3] Capture Visualisation. 2024. Capture. <https://www.capture.se/Products/Capture/Overview>
- [4] CAST Group of Companies. 2024. wysiwyg. <https://cast-soft.com/wysiwyg/>
- [5] Elizabeth Cha, Tushar Trehon, Lancelot Wathieu, Christian Wagner, Anurag Shukla, and Maja J Mataric. 2017. ModLight: Designing a modular light signaling tool for human-robot interaction. In *2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, Singapore, Singapore, 1654–1661. doi:10.1109/ICRA.2017.7989195
- [6] Ben Fry and Casey Reas. 2024. Processing. www.processing.org
- [7] William Gaver and Frances Gaver. 2023. Living with Light Touch: An Autoethnography of a Simple Communication Device in Long-Term Use. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3544548.3580807 event-place: Hamburg, Germany.
- [8] Eyal Ginosar and Jessica R. Cauchard. 2023. At First Light: Expressive Lights in Support of Drone-Initiated Communication. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (New York, NY, USA, 2023) (CHI '23). Association for Computing Machinery. doi:10.1145/3544548.3581062 event-place: Hamburg, Germany.
- [9] Chris Harrison, John Horstman, Gary Hsieh, and Scott Hudson. 2012. Unlocking the expressivity of point lights. In *Proceedings of the 2012 ACM annual conference on Human Factors in Computing Systems - CHI '12*. ACM Press, Austin, Texas, USA, 1683. doi:10.1145/2207676.2208296
- [10] Ann-Christin Hensch, Isabel Neumann, Matthias Beggiato, Josephine Halama, and Josef F Krems. 2019. Effects of a light-based communication approach as an external HMI for Automated Vehicles—a Wizard-of-Oz Study. *Transactions on Transport Sciences* 10, 2 (2019).
- [11] Marius Hoggemueller, Luke Hespanhol, and Martin Tomitsch. 2020. Stop and Smell the Chalk Flowers: A Robotic Probe for Investigating Urban Interaction with Physicalised Displays. (2020), 15.
- [12] Marius Hoggemueller, Martin Tomitsch, and Alexander Wiethoff. 2018. Understanding Artefact and Process Challenges for Designing Low-Res Lighting Displays. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3173574.3173833
- [13] Jan Hommes and Michael Kipp. 2024. Light Bridge: Improving social connectedness through ambient spatial interaction. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (New York, NY, USA, 2024) (DIS '24). Association for Computing Machinery, 1658–1665. doi:10.1145/3643834.3660689 event-place: Copenhagen, Denmark.
- [14] Sahiti Kunchay and Saeed Abdullah. 2021. Assessing Effectiveness and Interpretability of Light Behaviors in Smart Speakers. In *Proceedings of the 3rd Conference on Conversational User Interfaces* (New York, NY, USA, 2021) (CUI '21). Association for Computing Machinery. doi:10.1145/3469595.3469610 event-place: Bilbao (online), Spain.
- [15] Dolby Laboratories. 2025. *Dolby Atmos Specifications*. Technical Report. <https://professional.dolby.com/siteassets/cinema/dolby-audio-products/dolby-atmos-specifications.pdf> Accessed: 2025-01-17.
- [16] Dolby Laboratories. 2025. *Meet Dolby Atmos FlexConnect*. <https://www.dolby.com/technologies/dolby-atmos/dolby-atmos-flexconnect/> Accessed: 2025-01-17.
- [17] Mirjam Lanzer, Ina Koniakowsky, Mark Colley, and Martin Baumann. 2023. Interaction Effects of Pedestrian Behavior, Smartphone Distraction and External Communication of Automated Vehicles on Crossing and Gaze Behavior. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3544548.3581303 event-place: Hamburg, Germany.
- [18] MA Lighting International GmbH. 2024. GrandMA3. <https://www.malighting.com/special/grandma3-software/>
- [19] Martin Professional. 2021. Lighting Design for Concert Touring - Efficient Workflows and Approaches with Roland Greil - Webinar. https://www.youtube.com/watch?v=jjoHKCrI4_U Accessed: 2025-01-15.
- [20] Sichao Song and Seiji Yamada. 2018. Bioluminescence-inspired human-robot interaction: Designing expressive lights that affect human's willingness to interact with a robot. In *Proceedings of the 2018 ACM/IEEE international conference on human-robot interaction*. 224–232.
- [21] Kazunori Terada, Atsushi Yamauchi, and Akira Ito. 2012. Artificial emotion expression for a robot by dynamic color change. In *2012 IEEE RO-MAN: The 21st IEEE International Symposium on Robot and Human Interactive Communication*. IEEE, 314–321.
- [22] Vectorworks. 2024. Vectorworks Vision. <https://www.vectorworks.net/en-AU/vision>
- [23] Marc Wilbrink, Anna Schieben, and Michael Oehl. 2020. Reflecting the automated vehicle's perception and intention: Light-based interaction approaches

for on-board HMI in highly automated vehicles. In *Companion Proceedings of the 25th International Conference on Intelligent User Interfaces (IUI '20 Companion)*. Association for Computing Machinery, New York, NY, USA, 105–107. doi:10.1145/3379336.3381502 event-place: Cagliari, Italy.

- [24] Lei Zhang, Daekun Kim, Youjean Cho, Ava Robinson, Yu Jiang Tham, Rajan Vaish, and Andrés Monroy-Hernández. 2024. Jigsaw: Authoring Immersive Storytelling

Experiences with Augmented Reality and Internet of Things. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3613904.3642744 event-place: Honolulu, HI, USA.

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